

Instrumental Variables

Qingyuan Fang¹

¹Johns Hopkins University

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Definition

[Vector Notation] $\mathbf{x}$ is a $p \times 1$ random vector

$$y = \mathbf{x}'\beta + e \quad (1)$$

$$\mathbb{E}[\mathbf{x}e] \neq 0 \text{ (Endogeneity)}$$

The $k \times 1$ random vector $\mathbf{z}$ ($k \geq p$) is an **instrumental variable** for (1) if

$$\mathbb{E}[\mathbf{z}e] = 0 \text{ (Exogenous)}$$

$$\mathbb{E}[\mathbf{z}\mathbf{z}'] > 0$$

$$\text{rank}(\mathbb{E}[\mathbf{z}\mathbf{z}']) = p \text{ (Relevance)}.$$

Note: IV $\subset$ Generalized Method of Moments (GMM)

Moment Condition: $\mathbb{E}[\mathbf{z}(y - \mathbf{x}'\beta)] = 0$.

Estimators

$$\underbrace{Y}_{n \times 1} = \underbrace{X}_{n \times p} \underbrace{\beta}_{p \times 1} + \varepsilon$$

$$\underbrace{X}_{n \times p} = \underbrace{Z}_{n \times k} \underbrace{\Pi}_{k \times p} + v$$

- ① Two Stage Least Squares (2SLS) $\subset$ “estimation with generated regressors”
- Regress X on Z and get fitted values [multivariate regression (Chap11)]

$$\hat{X} = Z \underbrace{(Z'Z)^{-1} Z'X}_{\hat{\Pi}}$$

- Regress Y on $\hat{X}$

$$\hat{\beta}_{2SLS} = (\hat{X}'\hat{X})^{-1} \hat{X}'Y = (X'Z(Z'Z)^{-1}Z'X)^{-1} X'Z(Z'Z)^{-1} Z'Y$$

Estimators

$$\begin{aligned} Y &= X\beta + \varepsilon \\ &= Z \underbrace{\Pi\beta}_{\Lambda} + \underbrace{v\beta + \varepsilon}_w \\ X &= Z\Pi + v \end{aligned}$$

② Indirect Least Squares (ILS)

- First Stage: $\hat{\Pi} = (Z'Z)^{-1}Z'X$
- Reduced Form: $\hat{\Lambda} = (Z'Z)^{-1}Z'Y$
- $\hat{\beta}_{ILS} = \hat{\Pi}^{-1}\hat{\Lambda} = \hat{\beta}_{2SLS}$

Estimators

[Vector Notation]

$$y = \mathbf{x}'\beta + \underbrace{(\boldsymbol{\nu}'\rho + \xi)}_e$$
$$\mathbf{x}' = \mathbf{z}'\Pi + \boldsymbol{\nu}'$$

$\mathbb{E}[\mathbf{z}e] = 0$. $\mathbb{E}[\mathbf{z}\boldsymbol{\nu}'] = 0$. $\mathbb{E}[\boldsymbol{\nu}\xi] = 0$. Prove $\mathbb{E}[\mathbf{x}\xi] = 0$?

③ Control Function Regression (CR)

- Residual from First Stage: $\hat{\nu} = (I - (Z'Z)^{-1}Z')X$
- Regress Y on X and $\hat{\nu}$.

Estimators

[Vector Notation]

$$\begin{aligned}y &= \mathbf{x}'\beta + e \\&= \mathbf{z}' \underbrace{\Pi\beta}_{\Lambda} + \underbrace{\boldsymbol{\nu}'\beta + e}_w \\ \mathbf{x}' &= \mathbf{z}'\Pi + \boldsymbol{\nu}' \\ (w; \boldsymbol{\nu}) &\sim MNormal\end{aligned}$$

4 Limited Information Maximum Likelihood (LIML)

$$\hat{\beta}_{LIML} = \arg \max_{\beta} \frac{(Y - X\beta)'M_Z(Y - X\beta)}{(Y - X\beta)'P_Z(Y - X\beta)}$$

where $P_Z = Z(Z'Z)^{-1}Z'$ and $M_Z = I_n - P_Z$.

5 Split-Sample IV and JIVE

Tests

- Cragg and Donald test $\Rightarrow$ Instrument relevance
- Overidentification test $\Rightarrow$ Instrument validity
- Durbin-Wu-Hausman test $\Rightarrow$ Endogeneity

AJR (2001)

Acemoglu, D., Johnson, S., & Robinson, J. A. (2001). The colonial origins of comparative development: An empirical investigation. *American Economic Review*, 91(5), 1369–1401.

Question: Institutions and Property Rights $\Rightarrow$ Economic Performance?

Structural Equation:

$$\log y_i = \mu + \alpha R_i + X_i' \gamma + \varepsilon_i$$

y_i : income per capita. R_i : protection against expropriation risk.

Endogeneity

- **Reverse Causality:** Rich economies may choose or afford better institutions.
- **Omitted Variables:** Unobserved determinants of income differences naturally correlate with institutions (examples?).
- **Measurement Error:** R_i is constructed *ex post*, creating attenuation bias that may bias $\hat{\alpha}_{OLS}$ downward.

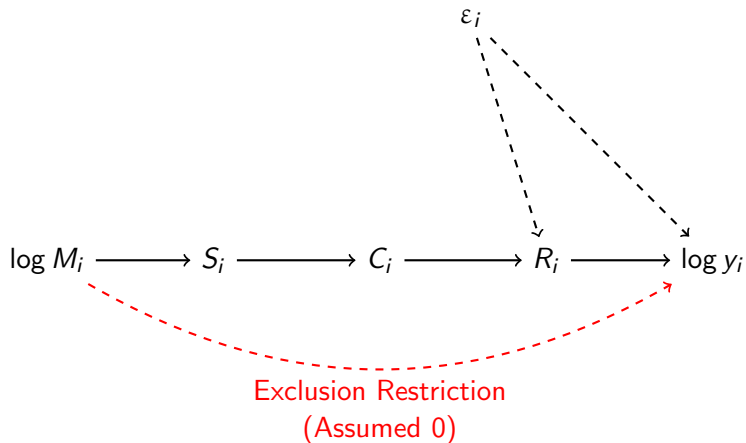
The IV Strategy: Settler Mortality

To estimate the impact of institutions, we need a source of exogenous variation for R_i . AJR propose potential European settler mortality ($\log M_i$).

The Proposed Mechanism:

- ① **Mortality** → **Settlements**: High disease environments deterred European settlement.
- ② **Settlements** → **Early Institutions**: Where Europeans settled, they built "Neo-Europes" (protecting property); where they could not, they built extractive states.
- ③ **Early Institutions** → **Current Institutions** Colonial states and institutions persisted even after independence.
- ④ **Current Institutions** → **Current Performance**

Causal DAG Design



- M_i : Settler mortality. S_i : Settlements. C_i : Early inst. R_i : Current inst. U_i :

Core IV Assumptions

For $\log M_i$ to be a valid instrument, two conditions must hold:

- **1. Relevance** $\text{cov}(\log M_i, R_i | X_i) \neq 0$
 - Settler mortality must strongly predict current institutional quality.
- **2. Exogeneity** $E[\varepsilon_i | \log M_i, X_i] = 0$
 - Conditional on controls, mortality rates > 100 years ago have *no effect* on current GDP per capita, other than through institutional development.
 - **The Narrative is Crucial.** An instrument must be supported by deep historical/institutional context. AJR relies heavily on detailed mortality records from historians like P.D. Curtin (Section I).

The 2SLS Estimator Framework

First Stage

$$R_i = \zeta + \beta \log M_i + X_i' \delta + v_i$$

- In the data, log settler mortality alone explains 27% of the differences in institutions we observe today.

Reduced Form:

$$\log y_i = \pi_0 + \pi_1 \log M_i + X_i' \pi_2 + u_i$$

Identification:

$$\alpha = \frac{\text{Cov}(\log y_i, \log M_i)}{\text{Cov}(R_i, \log M_i)} = \frac{\pi_1}{\beta}$$

Main Empirical Results

- **OLS Estimate:** $\hat{\alpha}_{OLS} = 0.52$.
- **2SLS Estimate:** $\hat{\alpha}_{2SLS} = 0.94$ (s.e. = 0.16).
- **Why is 2SLS > OLS?**
 - We usually expect OVB to bias OLS *upward* (richer places buy better institutions).
 - However, measurement error in complex "institutions" constructs creates severe *attenuation bias* (biasing OLS toward zero). Here, measurement error dominates.
- **Economic Magnitude:** Improving Nigeria's institutions to the level of Chile could lead to a 7-fold increase in Nigeria's income in the long run.

Test Alternative Channels: Section IV

The Threat to Validity: Settler mortality could correlate with the *current* disease environment, which might directly affect economic performance today.

Authors' Defense:

- Malaria and yellow fever were fatal to Europeans lacking immunity, but had limited effects on indigenous adults with developed immunity.
- *Robustness Check:* Control for fraction of population in endemic malaria area in 1994.
 $\hat{\alpha}_{2LS} = 0.69$ (s.e. = 0.25).
- Note: Controlling for endogenous malaria induces downward bias on the institutions coefficient (Appendix A).

Overidentification Tests

- AJR use early measures of early institutions (C_i) and European settlements (S_i) as additional instruments.
- **J-Test / Sargan Test Logic:** If Z_1 and Z_2 are both valid, they should yield statistically similar estimates of α .
- **Result:** AJR do not reject the overidentifying restrictions at the 5% level, supporting the claim that mortality affects income only through institutions.

Granular Instrumental Variables

Gabaix, X., & Koijen, R. S. J. (2024). Granular instrumental variables. *The Journal of Political Economy*, 132(7), 2274–2303.

- Countries $i = 1, \dots, n$ have world-market shares S_i , with $\sum_{i=1}^n S_i = 1$.
- **Demand by country i :**

$$y_{it} = \phi p_t + \eta_t + u_{it},$$

where p_t is the (world) price, η_t is a common demand shock, and u_{it} is idiosyncratic demand.

- **Aggregates:**

$$\begin{aligned} y_{st} &:= \sum_{i=1}^n S_i y_{it} = \phi p_t + \eta_t + u_{st}, & u_{st} &:= \sum_{i=1}^n S_i u_{it}, \\ y_{Et} &:= \frac{1}{n} \sum_{i=1}^n y_{it} = \phi p_t + \eta_t + u_{Et}, & u_{Et} &:= \frac{1}{n} \sum_{i=1}^n u_{it}. \end{aligned}$$

- **Supply:** $s_t = \lambda p_t + v_t$ and market clearing $y_{st} = s_t$.

Why OLS fails: price is endogenous

Market clearing implies

$$\phi p_t + \eta_t + u_{st} = \lambda p_t + v_t \quad \Rightarrow \quad p_t = \frac{\eta_t + u_{st} - v_t}{\lambda - \phi}.$$

- In the regression of global output on price (e.g. y_{st} on p_t), the regressor p_t loads on demand shocks (η_t, u_{st}).
- Hence $\mathbb{Cov}(p_t, \eta_t) \neq 0$ and $\mathbb{Cov}(p_t, u_{st}) \neq 0$ generically $\Rightarrow$ OLS mixes demand and supply.

Granular instrument and identification

Instrument (Gabaix–Koijen “granular”)

$$Z_t := y_{st} - y_{Et} = \sum_{i=1}^n \left(S_i - \frac{1}{n} \right) y_{it} = \sum_{i=1}^n \left(S_i - \frac{1}{n} \right) u_{it},$$

- **Supply slope** λ : IV regression $s_t = \lambda p_t + v_t$ using Z_t . Assumption: $E[u_{it}v_t] = 0 \ \forall i \Rightarrow E[Z_t v_t] = 0$.
- **Demand slope** ϕ : IV regression $y_{Et} = \phi p_t + \eta_t + u_{Et}$ using Z_t . Assumptions: $E[u_{it}\eta_t] = 0$, $E[u_{it}u_{jt}] = 0$ for $i \neq j$ (and finite variances) $\Rightarrow E[Z_t(\eta_t + u_{Et})] = 0$.
- **Relevance**: needs “granularity” ($S_i \neq 1/n$ for some i) and that u_{it} affects p_t via market clearing. “A few large firms, industries, or countries account for an important share of economic activity.”